CUSTOMER SERVICE REQUEST ANALYSIS

Final Data Analytics

Introduction

This project analyzes NYC 311 customer service request data. The main objective is to identify complaint patterns, analyze response times, clean the dataset, and visualize complaint distributions across different cities using Python data analytics libraries.

Objectives

- Understand the dataset
- Perform data cleaning
- Handle missing values
- Analyze complaint patterns
- Visualize complaints across cities
- Calculate complaint response times
- Perform statistical analysis
- Draw conclusions from the dataset

In [15]:
```
!pip install scipy
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting scipy
  Downloading scipy-1.17.1-cp314-cp314-win_amd64.whl.metadata (60 kB)
Requirement already satisfied: numpy<2.7,>=1.26.4 in c:\users\dell\appdata\roaming\p
ython\python314\site-packages (from scipy) (2.3.5)
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Installing collected packages: scipy
Successfully installed scipy-1.17.1
```

```
[notice] A new release of pip is available: 25.3 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [16]:

```python
# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import kruskal
```

In [17]:
```
# 2. Load Dataset
df = pd.read_csv(
    r"C:\Users\DELL\Downloads\311-service-requests-nyc\311_Service_Requests_from_20
    nrows=50000,
    low_memory=False
)
```

In [18]:
```
# 4. Dataset Columns
df.head()
```

Out[18]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Lo
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Stre
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Stre
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Stre

5 rows × 53 columns

In [19]:
```
# 4. Dataset Columns
print(df.columns)
```

```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

In [20]:

```
# 5. Dataset Shape
print(df.shape)
```

```
(50000, 53)
```

In [21]:

```
# 6. Dataset Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 53 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Unique Key                      50000 non-null  int64
 1   Created Date                    50000 non-null  object
 2   Closed Date                     49725 non-null  object
 3   Agency                          50000 non-null  object
 4   Agency Name                     50000 non-null  object
 5   Complaint Type                  50000 non-null  object
 6   Descriptor                      49283 non-null  object
 7   Location Type                   50000 non-null  object
 8   Incident Zip                    49679 non-null  float64
 9   Incident Address                44212 non-null  object
 10  Street Name                     44212 non-null  object
 11  Cross Street 1                  43457 non-null  object
 12  Cross Street 2                  43403 non-null  object
 13  Intersection Street 1           5733 non-null   object
 14  Intersection Street 2           5674 non-null   object
 15  Address Type                    49633 non-null  object
 16  City                            49678 non-null  object
 17  Landmark                        25 non-null     object
 18  Facility Type                   49730 non-null  object
 19  Status                          50000 non-null  object
 20  Due Date                        50000 non-null  object
 21  Resolution Description          50000 non-null  object
 22  Resolution Action Updated Date  49730 non-null  object
 23  Community Board                 50000 non-null  object
 24  Borough                         50000 non-null  object
 25  X Coordinate (State Plane)      49570 non-null  float64
 26  Y Coordinate (State Plane)      49570 non-null  float64
 27  Park Facility Name              50000 non-null  object
 28  Park Borough                    50000 non-null  object
 29  School Name                     50000 non-null  object
 30  School Number                   50000 non-null  object
 31  School Region                   50000 non-null  object
 32  School Code                     50000 non-null  object
 33  School Phone Number             50000 non-null  object
 34  School Address                  50000 non-null  object
 35  School City                     50000 non-null  object
 36  School State                    50000 non-null  object
 37  School Zip                      50000 non-null  object
 38  School Not Found                50000 non-null  object
 39  School or Citywide Complaint    0 non-null      float64
 40  Vehicle Type                    0 non-null      float64
 41  Taxi Company Borough            0 non-null      float64
 42  Taxi Pick Up Location           0 non-null      float64
 43  Bridge Highway Name             47 non-null     object
 44  Bridge Highway Direction        47 non-null     object
 45  Road Ramp                       38 non-null     object
 46  Bridge Highway Segment          38 non-null     object
 47  Garage Lot Name                 0 non-null      float64
 48  Ferry Direction                 0 non-null      float64
 49  Ferry Terminal Name             0 non-null      float64
 50  Latitude                        49570 non-null  float64
```

```
 51  Longitude                      49570 non-null  float64
 52  Location                       49570 non-null  object
dtypes: float64(12), int64(1), object(40)
memory usage: 20.2+ MB
```

In [22]:

```python
# 7. Statistical Summary
df.describe()
```

Out[22]:

	Unique Key	Incident Zip	X Coordinate (State Plane)	Y Coordinate (State Plane)	School or Citywide Complaint	Vehicle Type	Com Boro
count	5.000000e+04	49679.000000	4.957000e+04	49570.000000	0.0	0.0	
mean	3.213919e+07	10878.535659	1.004405e+06	201492.829453	NaN	NaN	
std	1.000917e+05	573.342338	2.237173e+04	29372.470018	NaN	NaN	
min	3.195831e+07	83.000000	9.133570e+05	121998.000000	NaN	NaN	
25%	3.205144e+07	10451.000000	9.905112e+05	180425.000000	NaN	NaN	
50%	3.214385e+07	11211.000000	1.002814e+06	199696.500000	NaN	NaN	
75%	3.222533e+07	11249.000000	1.019534e+06	218576.000000	NaN	NaN	
max	3.231065e+07	11697.000000	1.067154e+06	271391.000000	NaN	NaN	

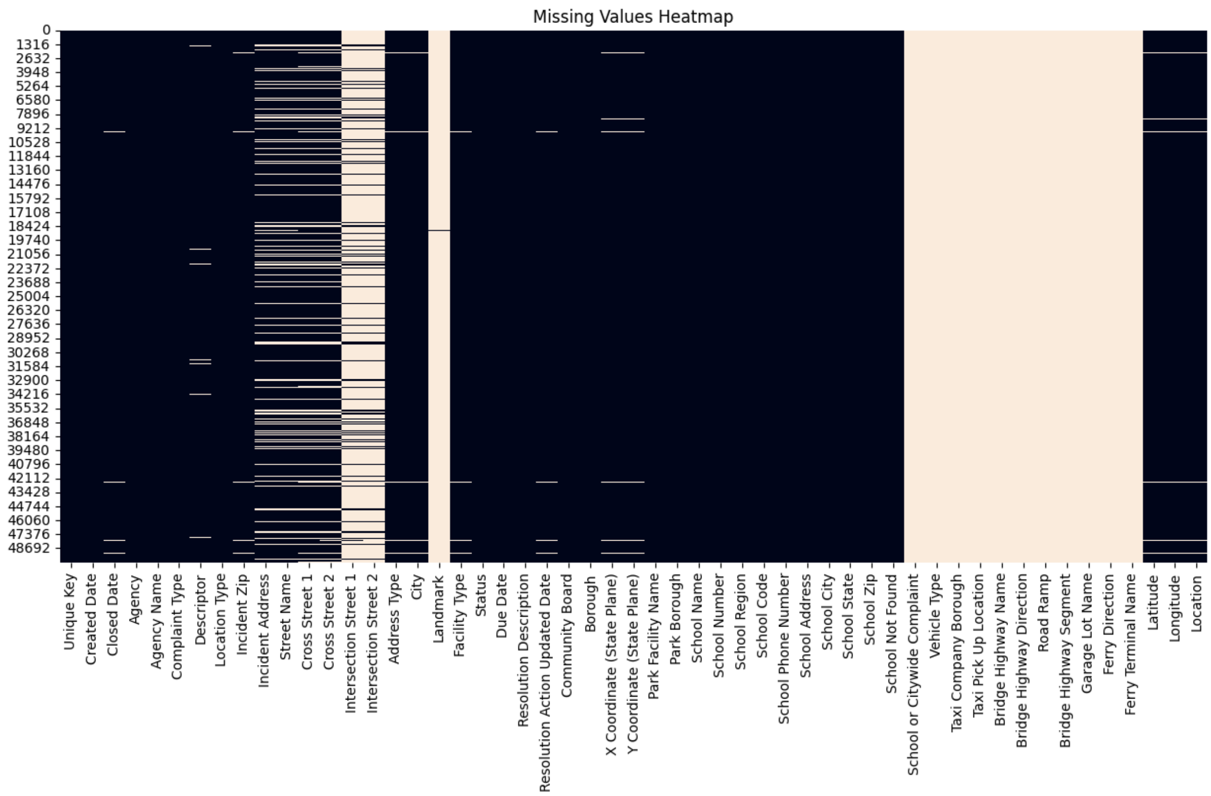
In [23]:

```python
# 8. Missing Value Analysis
print(df.isnull().sum())
```

```
Unique Key                            0
Created Date                          0
Closed Date                         275
Agency                                0
Agency Name                           0
Complaint Type                        0
Descriptor                          717
Location Type                         0
Incident Zip                        321
Incident Address                   5788
Street Name                        5788
Cross Street 1                     6543
Cross Street 2                     6597
Intersection Street 1             44267
Intersection Street 2             44326
Address Type                        367
City                                322
Landmark                          49975
Facility Type                       270
Status                                0
Due Date                              0
Resolution Description                0
Resolution Action Updated Date      270
Community Board                       0
Borough                               0
X Coordinate (State Plane)          430
Y Coordinate (State Plane)          430
Park Facility Name                    0
Park Borough                          0
School Name                           0
School Number                         0
School Region                         0
School Code                           0
School Phone Number                   0
School Address                        0
School City                           0
School State                          0
School Zip                            0
School Not Found                      0
School or Citywide Complaint      50000
Vehicle Type                      50000
Taxi Company Borough              50000
Taxi Pick Up Location             50000
Bridge Highway Name               49953
Bridge Highway Direction          49953
Road Ramp                         49962
Bridge Highway Segment            49962
Garage Lot Name                   50000
Ferry Direction                   50000
Ferry Terminal Name               50000
Latitude                            430
Longitude                           430
Location                            430
dtype: int64
```

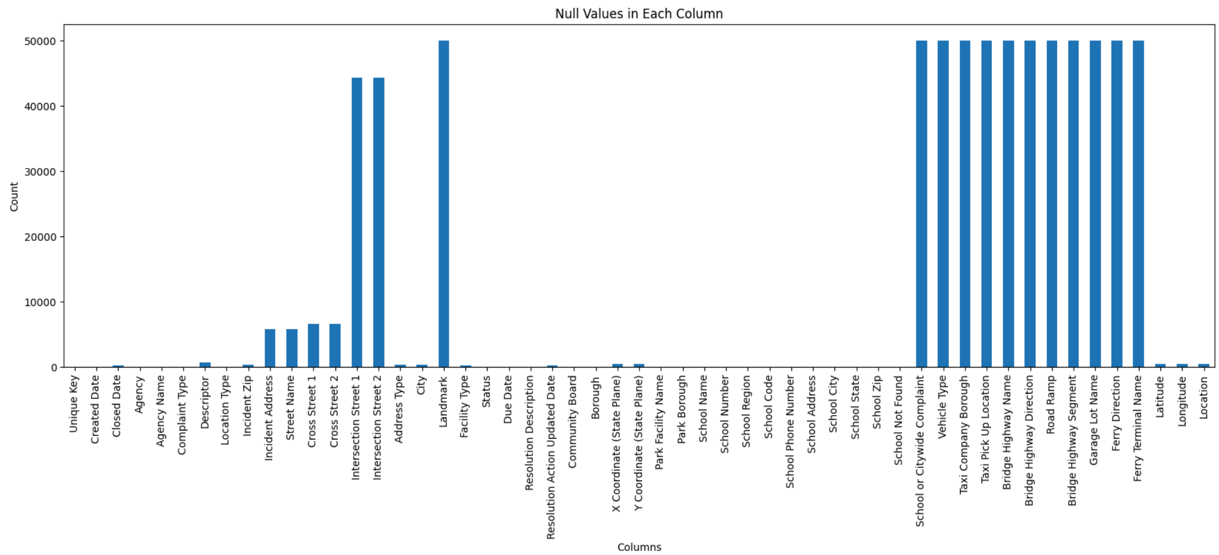
In [24]:

```python
plt.figure(figsize=(15,7))

sns.heatmap(
    df.isnull(),
    cbar=False
)

plt.title("Missing Values Heatmap")

plt.show()
```



In [26]:

```python
df.isnull().sum().plot(
    kind='bar',
    figsize=(20,6)
)

plt.title("Null Values in Each Column")

plt.xlabel("Columns")
plt.ylabel("Count")

plt.show()
```



Observation Several columns contain missing values. Some optional fields such as school information and taxi-related columns contain a high percentage of null values.

In [27]:
```
# 9. Data Cleaning
df = df[df['Closed Date'].notnull()]
```

In [29]:
```
df['Created Date'] = pd.to_datetime(df['Created Date'])
df['Closed Date'] = pd.to_datetime(df['Closed Date'])
```

In [30]:
```
df = df[
    df['Closed Date'] >= df['Created Date']
]
```

In [31]:
```
# 10. Request Closing Time Analysis
df['Request_Closing_Time'] = (
    df['Closed Date'] - df['Created Date']
)
```

In [32]:
```
df['Request_Closing_Time'] = (
    df['Request_Closing_Time']
    .dt.total_seconds()
)
```

In [33]:
```
df['Request_Closing_Time'].describe()
```

Out[33]:
```
count    4.972500e+04
mean     1.699324e+04
std      2.967123e+04
min      1.460000e+02
25%      4.697000e+03
50%      9.945000e+03
75%      1.971100e+04
max      2.078466e+06
Name: Request_Closing_Time, dtype: float64
```

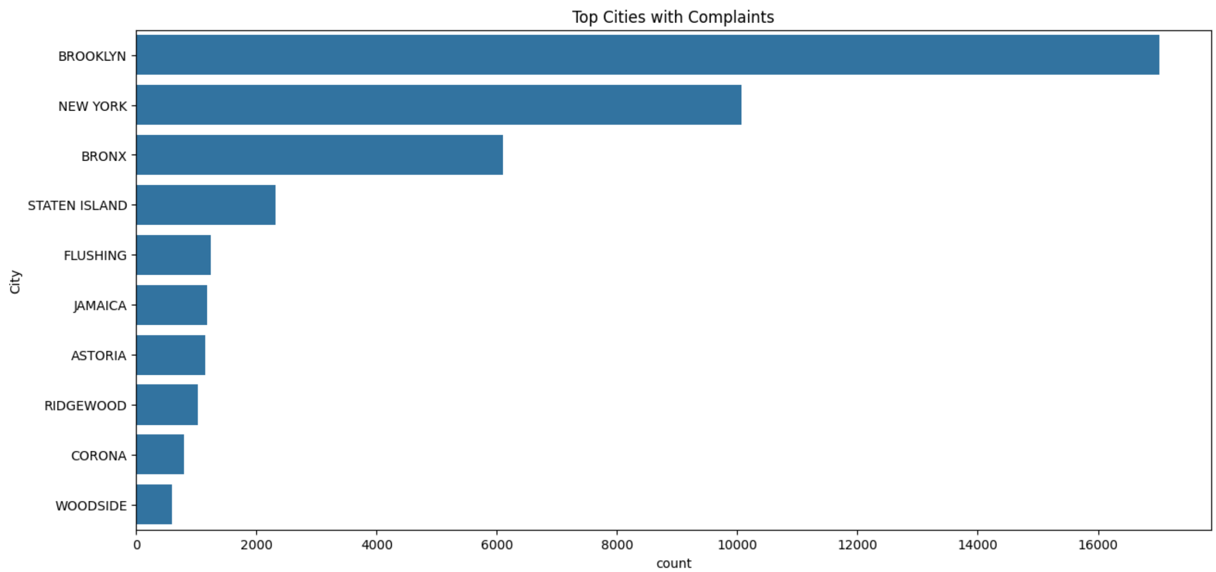
Observation The response time varies significantly across complaints. Some complaints are resolved quickly while others require longer processing time.

In [34]:
```
print(df['City'].isnull().sum())
```

61

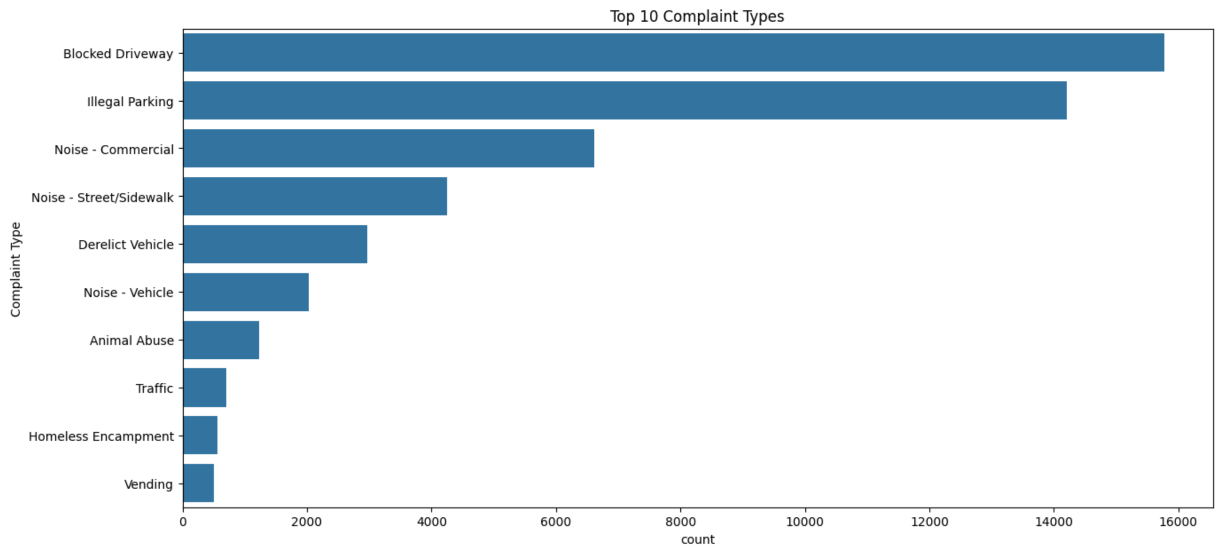
In [35]:
```
df['City'].fillna(
    "Unknown City",
    inplace=True
)
```

In [36]:
```
# 11. Complaints by City
plt.figure(figsize=(15,7))

sns.countplot(
    data=df,
    y='City',
    order=df['City'].value_counts().index[:10]
)

plt.title("Top Cities with Complaints")

plt.show()
```



Observation Brooklyn and New York City have the highest number of complaints.

In [37]:
```
plt.figure(figsize=(15,7))

sns.countplot(
    data=df,
    y='Complaint Type',
    order=df['Complaint Type'].value_counts().index[:10]
)

plt.title("Top 10 Complaint Types")

plt.show()
```



Observation Noise complaints are among the most common complaint categories in the dataset.

In [38]:
```
top10 = (
    df['Complaint Type']
    .value_counts()
    .head(10)
)

print(top10)
```

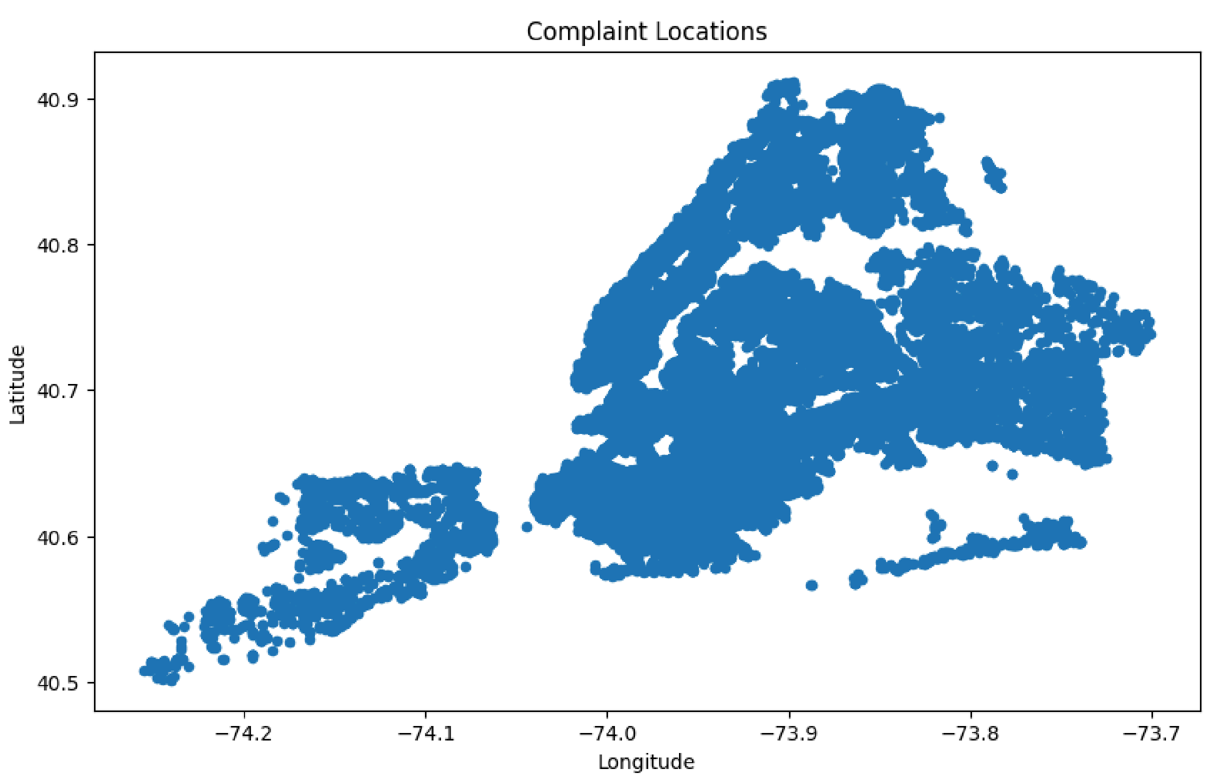
```
Complaint Type
Blocked Driveway           15771
Illegal Parking            14209
Noise - Commercial          6616
Noise - Street/Sidewalk     4253
Derelict Vehicle            2975
Noise - Vehicle             2034
Animal Abuse                1231
Traffic                      700
Homeless Encampment          566
Vending                      506
Name: count, dtype: int64
```

In [39]:
```
nyc = df[df['City'] == 'NEW YORK']

print(
    nyc['Complaint Type']
    .value_counts()
)
```

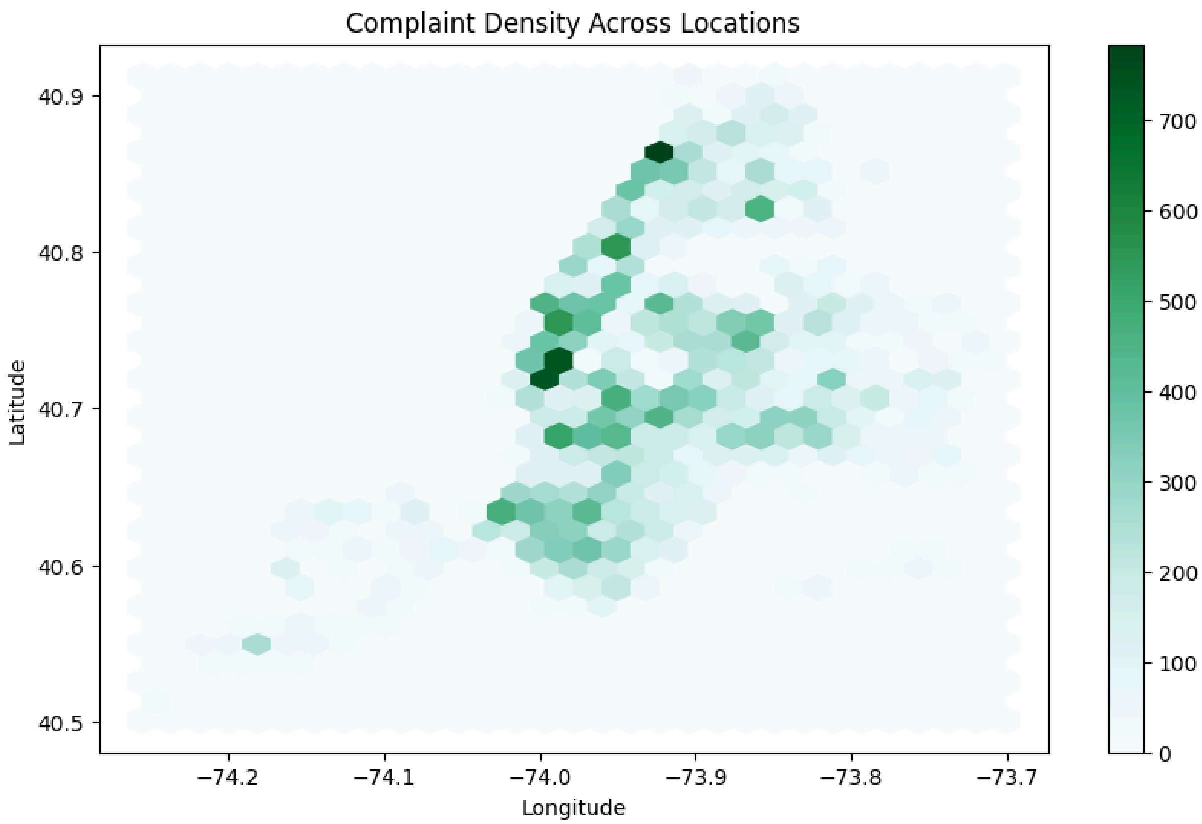
```
Complaint Type
Noise - Commercial           3023
Illegal Parking              2183
Noise - Street/Sidewalk      2021
Noise - Vehicle               866
Blocked Driveway              389
Homeless Encampment           375
Vending                       307
Traffic                       285
Animal Abuse                  272
Derelict Vehicle              125
Noise - Park                   84
Drinking                       35
Panhandling                    30
Urinating in Public            20
Noise - House of Worship       19
Bike/Roller/Skate Chronic      18
Posting Advertisement           8
Graffiti                        2
Disorderly Youth                2
Name: count, dtype: int64
```

In [40]:

```python
# 13. Complaint Location Analysis
df.plot.scatter(
    x='Longitude',
    y='Latitude',
    figsize=(10,6)
)

plt.title("Complaint Locations")

plt.show()
```



In [41]:
```
df.plot.hexbin(
    x='Longitude',
    y='Latitude',
    gridsize=30,
    figsize=(10,6)
)

plt.title("Complaint Density Across Locations")

plt.show()
```



In [42]:
```
# 14. Complaint Types Across Cities
city_complaints = pd.crosstab(
    df['City'],
    df['Complaint Type']
)

print(city_complaints)
```

Complaint Type	Animal Abuse	Bike/Roller/Skate Chronic	\
City			
ARVERNE	5	0	
ASTORIA	21	7	
BAYSIDE	5	0	
BELLEROSE	1	0	
BREEZY POINT	0	0	
BRONX	245	5	
BROOKLYN	325	8	
CAMBRIA HEIGHTS	0	0	
CENTRAL PARK	0	0	
COLLEGE POINT	3	0	
CORONA	2	0	
EAST ELMHURST	9	0	
ELMHURST	6	0	
FAR ROCKAWAY	21	0	
FLORAL PARK	1	0	
FLUSHING	32	0	
FOREST HILLS	11	1	
FRESH MEADOWS	6	0	
GLEN OAKS	0	0	
HOLLIS	2	0	
HOWARD BEACH	10	0	
JACKSON HEIGHTS	6	1	
JAMAICA	40	0	
KEW GARDENS	2	0	
LITTLE NECK	3	0	
LONG ISLAND CITY	4	0	
MASPETH	1	0	
MIDDLE VILLAGE	5	0	
NEW HYDE PARK	0	0	
NEW YORK	272	18	
OAKLAND GARDENS	3	0	
OZONE PARK	11	0	
QUEENS	0	0	
QUEENS VILLAGE	2	0	
REGO PARK	5	0	
RICHMOND HILL	2	0	
RIDGEWOOD	23	1	
ROCKAWAY PARK	7	0	
ROSEDALE	3	0	
SAINT ALBANS	4	0	
SOUTH OZONE PARK	16	0	
SOUTH RICHMOND HILL	2	0	
SPRINGFIELD GARDENS	4	0	
STATEN ISLAND	88	1	
SUNNYSIDE	1	1	
Unknown City	0	1	
WHITESTONE	1	1	
WOODHAVEN	11	0	
WOODSIDE	10	1	

Complaint Type	Blocked Driveway	Derelict Vehicle	Disorderly Youth	\
City				
ARVERNE	5	7	0	
ASTORIA	546	55	0	

BAYSIDE	60	31	0
BELLEROSE	17	7	0
BREEZY POINT	1	0	0
BRONX	2597	333	2
BROOKLYN	5999	920	4
CAMBRIA HEIGHTS	25	15	0
CENTRAL PARK	0	0	0
COLLEGE POINT	105	20	0
CORONA	590	7	0
EAST ELMHURST	262	24	0
ELMHURST	285	18	0
FAR ROCKAWAY	45	24	0
FLORAL PARK	2	4	0
FLUSHING	622	78	0
FOREST HILLS	129	6	0
FRESH MEADOWS	122	44	0
GLEN OAKS	5	4	0
HOLLIS	80	29	0
HOWARD BEACH	35	20	0
JACKSON HEIGHTS	127	3	0
JAMAICA	577	151	0
KEW GARDENS	60	4	0
LITTLE NECK	25	11	0
LONG ISLAND CITY	135	28	0
MASPETH	141	80	0
MIDDLE VILLAGE	67	62	0
NEW HYDE PARK	6	1	0
NEW YORK	389	125	2
OAKLAND GARDENS	30	7	0
OZONE PARK	224	78	0
QUEENS	0	0	0
QUEENS VILLAGE	119	60	0
REGO PARK	125	16	0
RICHMOND HILL	138	26	0
RIDGEWOOD	377	47	0
ROCKAWAY PARK	12	1	0
ROSEDALE	28	26	0
SAINT ALBANS	57	28	0
SOUTH OZONE PARK	168	46	0
SOUTH RICHMOND HILL	300	51	0
SPRINGFIELD GARDENS	53	30	0
STATEN ISLAND	478	305	3
SUNNYSIDE	58	1	0
Unknown City	9	6	0
WHITESTONE	40	27	0
WOODHAVEN	198	55	0
WOODSIDE	298	54	0

Complaint Type	Drinking	Graffiti	Homeless Encampment	\
City				
ARVERNE	0	0	1	
ASTORIA	4	0	4	
BAYSIDE	0	2	1	
BELLEROSE	0	0	0	
BREEZY POINT	0	0	0	
BRONX	13	2	27	

BROOKLYN	19	3	102
CAMBRIA HEIGHTS	0	0	1
CENTRAL PARK	0	0	0
COLLEGE POINT	0	0	0
CORONA	4	0	1
EAST ELMHURST	0	0	0
ELMHURST	1	0	13
FAR ROCKAWAY	0	0	2
FLORAL PARK	1	0	0
FLUSHING	9	1	1
FOREST HILLS	0	0	2
FRESH MEADOWS	0	0	0
GLEN OAKS	0	0	0
HOLLIS	0	0	0
HOWARD BEACH	0	0	0
JACKSON HEIGHTS	1	0	1
JAMAICA	0	0	7
KEW GARDENS	0	0	0
LITTLE NECK	0	0	0
LONG ISLAND CITY	1	1	0
MASPETH	1	0	1
MIDDLE VILLAGE	0	0	1
NEW HYDE PARK	0	0	0
NEW YORK	35	2	375
OAKLAND GARDENS	0	0	0
OZONE PARK	3	0	1
QUEENS	0	0	0
QUEENS VILLAGE	0	0	3
REGO PARK	2	0	0
RICHMOND HILL	1	0	2
RIDGEWOOD	2	0	1
ROCKAWAY PARK	1	0	0
ROSEDALE	1	0	0
SAINT ALBANS	0	0	0
SOUTH OZONE PARK	0	0	0
SOUTH RICHMOND HILL	2	0	0
SPRINGFIELD GARDENS	0	0	0
STATEN ISLAND	60	0	17
SUNNYSIDE	1	0	0
Unknown City	0	0	1
WHITESTONE	0	0	0
WOODHAVEN	0	0	1
WOODSIDE	5	1	0

Complaint Type	Illegal Fireworks	Illegal Parking	Noise - Commercial \
City			
ARVERNE	0	15	2
ASTORIA	0	251	184
BAYSIDE	0	93	4
BELLEROSE	0	7	4
BREEZY POINT	0	0	0
BRONX	1	1475	501
BROOKLYN	1	5434	2061
CAMBRIA HEIGHTS	0	14	3
CENTRAL PARK	0	0	0
COLLEGE POINT	0	42	12

```
CORONA                                    0              137                     18
EAST ELMHURST                             0              151                     10
ELMHURST                                  0              142                     17
FAR ROCKAWAY                              0               59                      6
FLORAL PARK                               0                8                      0
FLUSHING                                  1              408                     27
FOREST HILLS                              0              100                     28
FRESH MEADOWS                             0              260                      3
GLEN OAKS                                 0                8                     23
HOLLIS                                    0               33                      4
HOWARD BEACH                              0               66                     13
JACKSON HEIGHTS                           0               36                     82
JAMAICA                                   0              228                     75
KEW GARDENS                               0               30                     27
LITTLE NECK                               0               58                      4
LONG ISLAND CITY                          0              145                     55
MASPETH                                   0              180                     12
MIDDLE VILLAGE                            0              181                      2
NEW HYDE PARK                             0                2                      0
NEW YORK                                  0             2183                   3023
OAKLAND GARDENS                           0               50                      0
OZONE PARK                                0               94                     13
QUEENS                                    0                1                      1
QUEENS VILLAGE                            0              100                     14
REGO PARK                                 0              131                     25
RICHMOND HILL                             0               62                     44
RIDGEWOOD                                 0              423                     92
ROCKAWAY PARK                             0               30                      3
ROSEDALE                                  0               41                      1
SAINT ALBANS                              0               44                      6
SOUTH OZONE PARK                          0               58                      8
SOUTH RICHMOND HILL                       0               72                     37
SPRINGFIELD GARDENS                       0               44                      0
STATEN ISLAND                             0              876                     92
SUNNYSIDE                                 0               23                     15
Unknown City                              0               29                     10
WHITESTONE                                0               91                      2
WOODHAVEN                                 0              138                     23
WOODSIDE                                  0              156                     30

Complaint Type       Noise - House of Worship  Noise - Park  \
City
ARVERNE                                     0             0
ASTORIA                                     2             2
BAYSIDE                                     0             0
BELLEROSE                                   0             0
BREEZY POINT                                0             0
BRONX                                       8            13
BROOKLYN                                   44            60
CAMBRIA HEIGHTS                             0             0
CENTRAL PARK                                0             0
COLLEGE POINT                               0             0
CORONA                                      2             0
EAST ELMHURST                               1             0
ELMHURST                                    0             2
FAR ROCKAWAY                                0             0
```

FLORAL PARK	0	0
FLUSHING	0	1
FOREST HILLS	0	1
FRESH MEADOWS	0	1
GLEN OAKS	0	3
HOLLIS	1	0
HOWARD BEACH	0	0
JACKSON HEIGHTS	1	1
JAMAICA	0	1
KEW GARDENS	0	0
LITTLE NECK	0	0
LONG ISLAND CITY	0	1
MASPETH	0	0
MIDDLE VILLAGE	0	0
NEW HYDE PARK	0	0
NEW YORK	19	84
OAKLAND GARDENS	0	0
OZONE PARK	0	0
QUEENS	0	0
QUEENS VILLAGE	0	0
REGO PARK	0	1
RICHMOND HILL	0	0
RIDGEWOOD	0	4
ROCKAWAY PARK	0	0
ROSEDALE	0	0
SAINT ALBANS	0	0
SOUTH OZONE PARK	1	0
SOUTH RICHMOND HILL	0	0
SPRINGFIELD GARDENS	0	0
STATEN ISLAND	2	1
SUNNYSIDE	0	1
Unknown City	0	0
WHITESTONE	0	1
WOODHAVEN	2	0
WOODSIDE	0	6

Complaint Type	Noise - Street/Sidewalk	Noise - Vehicle	Panhandling \
City			
ARVERNE	2	1	0
ASTORIA	48	19	0
BAYSIDE	3	2	0
BELLEROSE	2	0	0
BREEZY POINT	1	0	0
BRONX	585	201	1
BROOKLYN	1147	627	9
CAMBRIA HEIGHTS	2	10	0
CENTRAL PARK	15	0	0
COLLEGE POINT	8	28	0
CORONA	18	8	0
EAST ELMHURST	6	6	0
ELMHURST	14	4	0
FAR ROCKAWAY	12	6	0
FLORAL PARK	0	0	0
FLUSHING	30	10	0
FOREST HILLS	8	10	2
FRESH MEADOWS	3	13	0

GLEN OAKS	1	1	0
HOLLIS	1	2	0
HOWARD BEACH	0	0	1
JACKSON HEIGHTS	11	4	0
JAMAICA	18	24	0
KEW GARDENS	2	2	0
LITTLE NECK	0	1	0
LONG ISLAND CITY	15	17	0
MASPETH	53	4	0
MIDDLE VILLAGE	5	14	0
NEW HYDE PARK	0	0	0
NEW YORK	2021	866	30
OAKLAND GARDENS	2	3	0
OZONE PARK	3	6	1
QUEENS	0	0	0
QUEENS VILLAGE	6	15	1
REGO PARK	9	10	0
RICHMOND HILL	21	7	0
RIDGEWOOD	33	19	0
ROCKAWAY PARK	12	2	0
ROSEDALE	2	1	0
SAINT ALBANS	8	3	0
SOUTH OZONE PARK	3	12	0
SOUTH RICHMOND HILL	8	4	0
SPRINGFIELD GARDENS	0	4	0
STATEN ISLAND	74	39	4
SUNNYSIDE	10	6	0
Unknown City	5	0	0
WHITESTONE	5	1	0
WOODHAVEN	8	5	0
WOODSIDE	13	17	0

Complaint Type	Posting Advertisement	Traffic	Urinating in Public \
City			
ARVERNE	0	0	1
ASTORIA	0	8	1
BAYSIDE	0	0	0
BELLEROSE	0	1	0
BREEZY POINT	0	0	0
BRONX	0	50	3
BROOKLYN	12	161	12
CAMBRIA HEIGHTS	0	0	0
CENTRAL PARK	0	0	0
COLLEGE POINT	0	2	0
CORONA	0	1	1
EAST ELMHURST	0	3	0
ELMHURST	0	4	2
FAR ROCKAWAY	0	0	0
FLORAL PARK	0	0	0
FLUSHING	1	8	1
FOREST HILLS	0	19	0
FRESH MEADOWS	0	4	0
GLEN OAKS	0	0	0
HOLLIS	0	0	0
HOWARD BEACH	0	1	0
JACKSON HEIGHTS	0	2	0

JAMAICA	1	50	2
KEW GARDENS	0	2	0
LITTLE NECK	0	3	0
LONG ISLAND CITY	0	9	0
MASPETH	0	14	1
MIDDLE VILLAGE	0	2	0
NEW HYDE PARK	0	0	0
NEW YORK	8	285	20
OAKLAND GARDENS	0	1	0
OZONE PARK	0	2	0
QUEENS	0	0	0
QUEENS VILLAGE	0	4	2
REGO PARK	0	1	0
RICHMOND HILL	1	0	0
RIDGEWOOD	0	1	2
ROCKAWAY PARK	0	0	0
ROSEDALE	0	6	0
SAINT ALBANS	0	1	0
SOUTH OZONE PARK	0	5	0
SOUTH RICHMOND HILL	0	2	0
SPRINGFIELD GARDENS	0	3	0
STATEN ISLAND	235	36	2
SUNNYSIDE	0	1	0
Unknown City	0	0	0
WHITESTONE	0	1	0
WOODHAVEN	0	0	0
WOODSIDE	0	7	1

Complaint Type	Vending
City	
ARVERNE	0
ASTORIA	2
BAYSIDE	0
BELLEROSE	0
BREEZY POINT	0
BRONX	41
BROOKLYN	81
CAMBRIA HEIGHTS	0
CENTRAL PARK	0
COLLEGE POINT	0
CORONA	14
EAST ELMHURST	1
ELMHURST	7
FAR ROCKAWAY	0
FLORAL PARK	0
FLUSHING	4
FOREST HILLS	2
FRESH MEADOWS	0
GLEN OAKS	0
HOLLIS	0
HOWARD BEACH	0
JACKSON HEIGHTS	15
JAMAICA	3
KEW GARDENS	0
LITTLE NECK	0
LONG ISLAND CITY	3

```
MASPETH                    0
MIDDLE VILLAGE             0
NEW HYDE PARK              0
NEW YORK                 307
OAKLAND GARDENS            0
OZONE PARK                 0
QUEENS                     0
QUEENS VILLAGE             0
REGO PARK                  0
RICHMOND HILL              7
RIDGEWOOD                  1
ROCKAWAY PARK              1
ROSEDALE                   3
SAINT ALBANS               0
SOUTH OZONE PARK           0
SOUTH RICHMOND HILL        4
SPRINGFIELD GARDENS        0
STATEN ISLAND              3
SUNNYSIDE                  5
Unknown City               0
WHITESTONE                 0
WOODHAVEN                  0
WOODSIDE                   2
```

In [43]:

```
df_new = city_complaints.transpose()

print(df_new)
```

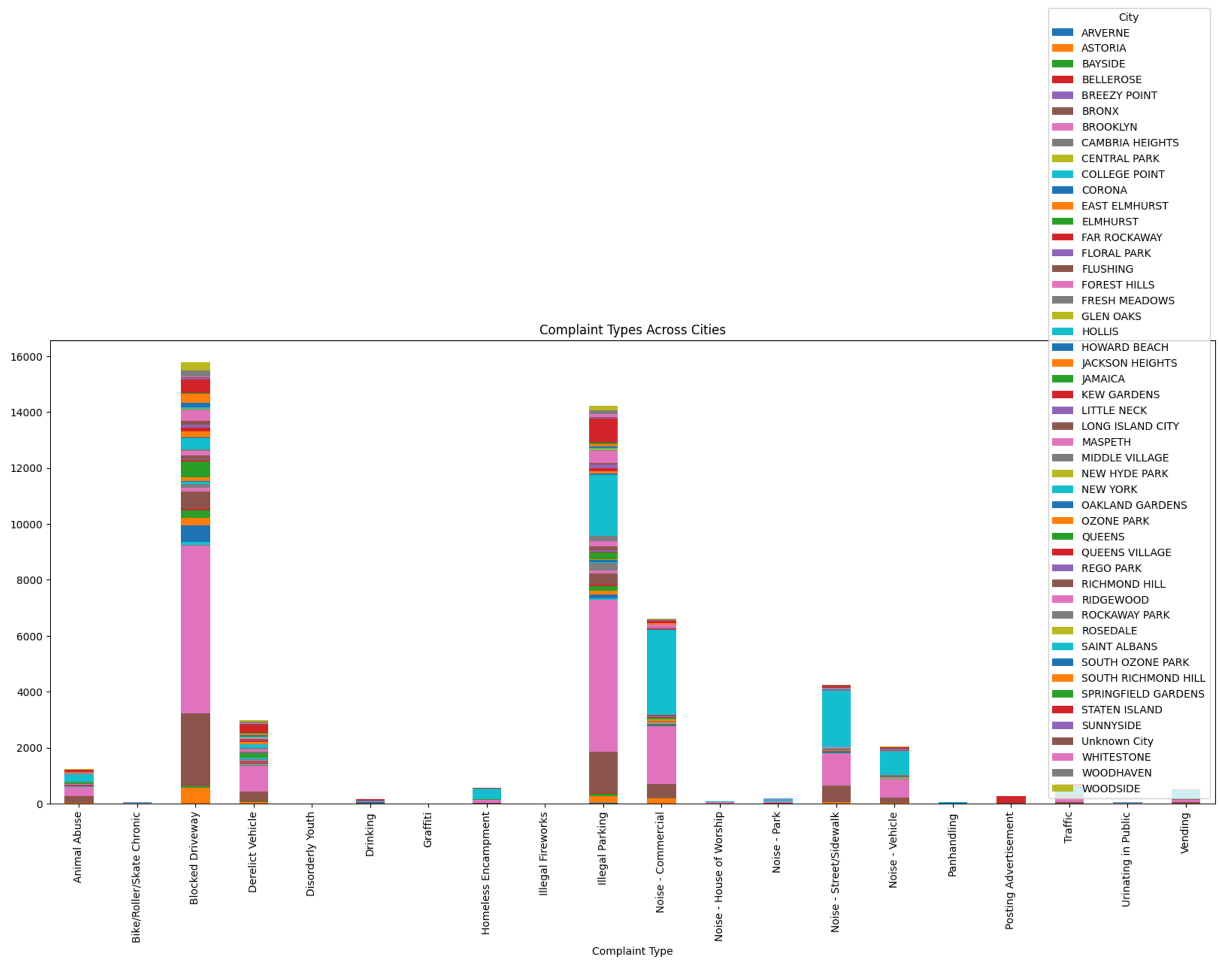
```
City                       ARVERNE  ASTORIA  BAYSIDE  BELLEROSE  BREEZY POINT  \
Complaint Type
Animal Abuse                     5       21        5          1             0
Bike/Roller/Skate Chronic        0        7        0          0             0
Blocked Driveway                 5      546       60         17             1
Derelict Vehicle                 7       55       31          7             0
Disorderly Youth                 0        0        0          0             0
Drinking                         0        4        0          0             0
Graffiti                         0        0        2          0             0
Homeless Encampment              1        4        1          0             0
Illegal Fireworks                0        0        0          0             0
Illegal Parking                 15      251       93          7             0
Noise - Commercial               2      184        4          4             0
Noise - House of Worship         0        2        0          0             0
Noise - Park                     0        2        0          0             0
Noise - Street/Sidewalk          2       48        3          2             1
Noise - Vehicle                  1       19        2          0             0
Panhandling                      0        0        0          0             0
Posting Advertisement            0        0        0          0             0
Traffic                          0        8        0          1             0
Urinating in Public              1        1        0          0             0
Vending                          0        2        0          0             0

City                       BRONX  BROOKLYN  CAMBRIA HEIGHTS  CENTRAL PARK  \
Complaint Type
Animal Abuse                 245       325                0             0
Bike/Roller/Skate Chronic      5         8                0             0
Blocked Driveway            2597      5999               25             0
Derelict Vehicle             333       920               15             0
Disorderly Youth               2         4                0             0
Drinking                      13        19                0             0
Graffiti                       2         3                0             0
Homeless Encampment           27       102                1             0
Illegal Fireworks              1         1                0             0
Illegal Parking             1475      5434               14             0
Noise - Commercial           501      2061                3             0
Noise - House of Worship       8        44                0             0
Noise - Park                  13        60                0             0
Noise - Street/Sidewalk      585      1147                2            15
Noise - Vehicle              201       627               10             0
Panhandling                    1         9                0             0
Posting Advertisement          0        12                0             0
Traffic                       50       161                0             0
Urinating in Public            3        12                0             0
Vending                       41        81                0             0

City                       COLLEGE POINT  ...  SAINT ALBANS  SOUTH OZONE PARK  \
Complaint Type                            ...
Animal Abuse                           3  ...             4                16
Bike/Roller/Skate Chronic              0  ...             0                 0
Blocked Driveway                     105  ...            57               168
Derelict Vehicle                      20  ...            28                46
Disorderly Youth                       0  ...             0                 0
Drinking                               0  ...             0                 0
Graffiti                               0  ...             0                 0
Homeless Encampment                    0  ...             0                 0
```

```
Illegal Fireworks                    0  ...            0                  0
Illegal Parking                     42  ...           44                 58
Noise - Commercial                  12  ...            6                  8
Noise - House of Worship             0  ...            0                  1
Noise - Park                         0  ...            0                  0
Noise - Street/Sidewalk              8  ...            8                  3
Noise - Vehicle                     28  ...            3                 12
Panhandling                          0  ...            0                  0
Posting Advertisement                0  ...            0                  0
Traffic                              2  ...            1                  5
Urinating in Public                  0  ...            0                  0
Vending                              0  ...            0                  0

City                       SOUTH RICHMOND HILL  SPRINGFIELD GARDENS  \
Complaint Type
Animal Abuse                                 2                    4
Bike/Roller/Skate Chronic                    0                    0
Blocked Driveway                           300                   53
Derelict Vehicle                            51                   30
Disorderly Youth                             0                    0
Drinking                                     2                    0
Graffiti                                     0                    0
Homeless Encampment                          0                    0
Illegal Fireworks                            0                    0
Illegal Parking                             72                   44
Noise - Commercial                          37                    0
Noise - House of Worship                     0                    0
Noise - Park                                 0                    0
Noise - Street/Sidewalk                      8                    0
Noise - Vehicle                              4                    4
Panhandling                                  0                    0
Posting Advertisement                        0                    0
Traffic                                      2                    3
Urinating in Public                          0                    0
Vending                                      4                    0

City                       STATEN ISLAND  SUNNYSIDE  Unknown City  WHITESTONE  \
Complaint Type
Animal Abuse                          88          1             0           1
Bike/Roller/Skate Chronic              1          1             1           1
Blocked Driveway                     478         58             9          40
Derelict Vehicle                     305          1             6          27
Disorderly Youth                       3          0             0           0
Drinking                              60          1             0           0
Graffiti                               0          0             0           0
Homeless Encampment                   17          0             1           0
Illegal Fireworks                      0          0             0           0
Illegal Parking                      876         23            29          91
Noise - Commercial                    92         15            10           2
Noise - House of Worship               2          0             0           0
Noise - Park                           1          1             0           1
Noise - Street/Sidewalk               74         10             5           5
Noise - Vehicle                       39          6             0           1
Panhandling                            4          0             0           0
Posting Advertisement                235          0             0           0
Traffic                               36          1             0           1
```

```
Urinating in Public                        2          0             0            0
Vending                                    3          5             0            0

City                       WOODHAVEN  WOODSIDE
Complaint Type
Animal Abuse                      11        10
Bike/Roller/Skate Chronic          0         1
Blocked Driveway                 198       298
Derelict Vehicle                  55        54
Disorderly Youth                   0         0
Drinking                           0         5
Graffiti                           0         1
Homeless Encampment                1         0
Illegal Fireworks                  0         0
Illegal Parking                  138       156
Noise - Commercial                23        30
Noise - House of Worship           2         0
Noise - Park                       0         6
Noise - Street/Sidewalk            8        13
Noise - Vehicle                    5        17
Panhandling                        0         0
Posting Advertisement              0         0
Traffic                            0         7
Urinating in Public                0         1
Vending                            0         2

[20 rows x 49 columns]
```

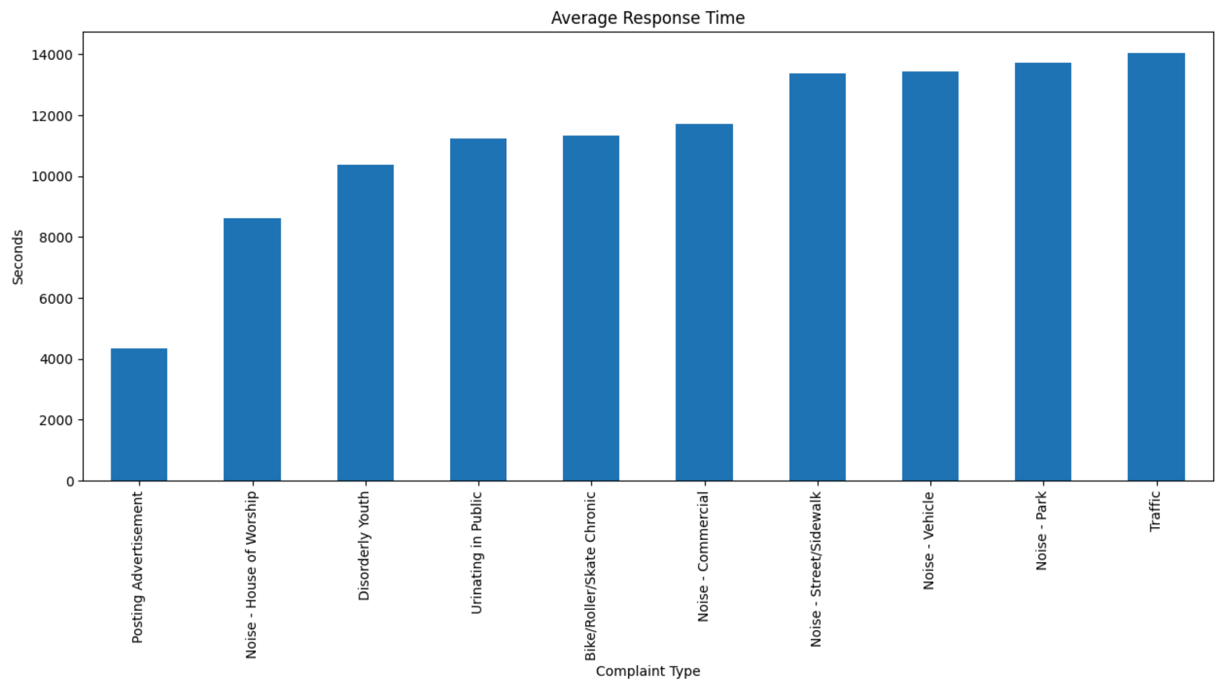
In [44]:

```
df_new.plot(
    kind='bar',
    stacked=True,
    figsize=(20,8)
)

plt.title("Complaint Types Across Cities")

plt.show()
```



In [46]:
```python
# 15. Average Response Time Analysis
avg_time = df.groupby(
    'Complaint Type'
)['Request_Closing_Time'].mean()
```

In [47]:
```python
avg_time.sort_values().head(10).plot(
    kind='bar',
    figsize=(15,6)
)

plt.title("Average Response Time")

plt.ylabel("Seconds")

plt.show()
```



In [48]:

```python
# 16. Statistical Analysis
complaints = df[
    'Complaint Type'
].dropna().unique()[:3]

data = []

for c in complaints:

    temp = df[
        df['Complaint Type'] == c
    ]['Request_Closing_Time']

    data.append(temp)

stat, p = kruskal(*data)

print("P-value:", p)

if p < 0.05:
    print("Reject H0")
else:
    print("Fail to Reject H0")
```

```
P-value: 1.2108699668454152e-112
Reject H0
```

In [49]:

```python
complaints = df[
    'Complaint Type'
].dropna().unique()[:3]

data = []

for c in complaints:
```

```
    temp = df[
        df['Complaint Type'] == c
    ]['Request_Closing_Time']

    data.append(temp)

stat, p = kruskal(*data)

print("P-value:", p)

if p < 0.05:
    print("Reject H0")
else:
    print("Fail to Reject H0")
```

```
P-value: 1.2108699668454152e-112
Reject H0
```

Interpretation

If p-value is less than 0.05, response times differ significantly across complaint categories. Otherwise, complaint response distributions are similar.

Conclusion

The analysis revealed important complaint patterns across NYC. Noise-related complaints were among the most common complaint categories. Brooklyn and New York City recorded the highest number of complaints. Response times varied across complaint categories, and statistical analysis helped identify significant differences in complaint resolution times.

In [ ]:

```

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